

2026, SSC Geometry Board Exam Solution

Time: 2 Hours

Mathematics (Part – II)

Max. Marks: 40

1. (A) For each of the following sub-questions four alternative answers are given. Choose the correct alternative and write its alphabet : [4]

(1) Out of the following which is the Pythagorean triplet ?

(A) (1, 5, 10)

(B) (3, 4, 5)

(C) (2, 2, 2)

(D) (5, 5, 2)

Ans: (B) **(3, 4, 5)**

EXPLANATION:

$$3^2 + 4^2 = 9 + 16 = 25 \dots\dots (i)$$

$$5^2 = 25 \dots\dots (ii)$$

$$\therefore 3^2 + 4^2 = 5^2 \dots\dots [\text{From (i) and (ii)}]$$

$\therefore$ (3, 4, 5) is a Pythagorean triplet.

(2) Two circles having radii 4 cm and 5 cm touch each other internally. Then the distance between the centres is

(A) 4 cm

(B) 5 cm

(C) 1 cm

(D) 9 cm

Ans: (C) **1 cm**

EXPLANATION:

The distance between the centres of the circles touching internally is equal to the difference of their radii.

$$\therefore \text{Distance between centres} = 5 - 4 = 1 \text{ cm}$$

(3) Out of the following points lies to the right of the origin on X-axis.

(A) $(-2, 0)$

(B) $(0, 2)$

(C) $(2, 3)$

(D) $(2, 0)$

Ans: (D) **$(2, 0)$**

EXPLANATION:

The y co-ordinate of every point on the X-axis is 0. As the x co-ordinate 2 is positive, the point $(2, 0)$ lies on the X-axis to the right of the origin.

(4) If the side of the cube is 5 cm, then the volume of that cube is

(A) 10 cm^3

(B) 100 cm^3

(C) 125 cm^3

(D) 25 cm^3

Ans: (C) **125 cm^3**

EXPLANATION:

Volume of cube = $(\text{side})^3 = 5^3 = 125 \text{ cm}^3$

(B) Solve the following sub-questions :

[4]

(1) The ratio of corresponding sides of similar triangles is 3 : 5, then find the ratio of their areas.

Ans:

Let $\triangle ABC \sim \triangle PQR$

$$\frac{A(\triangle ABC)}{A(\triangle PQR)} = \frac{AB^2}{PQ^2} \quad \dots \text{(Theorem of areas of similar triangles)}$$

$$\text{But } \frac{AB}{PQ} = \frac{3}{5} \quad \dots \text{(Given)}$$

$$\therefore \frac{A(\triangle ABC)}{A(\triangle PQR)} = \frac{AB^2}{PQ^2} = \frac{3^2}{5^2} = \frac{9}{25}$$

$\therefore$ Ratio of areas of similar triangles is **9 : 25**.

(2) Find the side of a square if its diagonal is $10\sqrt{2}$ cm.

Ans:

Diagonal = $10\sqrt{2}$ cm

For square, Side = $\frac{1}{\sqrt{2}} \times \text{Diagonal}$

$$\therefore \text{Side} = \frac{1}{\sqrt{2}} \times 10\sqrt{2} = \mathbf{10}$$

$\therefore$ Side of the square is 10 cm.

(3) If the measure of minor arc is 70° , then find the measure of corresponding major arc.

Ans:

Measure of minor arc = 70°

Measure of major arc = $360^\circ - \text{Measure of minor arc}$... (Measure of a circle is 360°)

$$= 360^\circ - 70^\circ = \mathbf{290^\circ}$$

$\therefore$ Measure of corresponding major arc is 290° .

(4) Angle made by the line with the positive direction of X-axis is 45° , find the slope of line.

Ans:

$$\theta = 45^\circ$$

$$\text{Slope} = \tan \theta = \tan 45^\circ$$

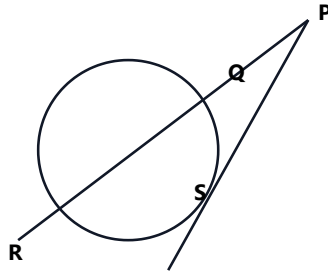
$$\therefore \text{Slope} = \mathbf{1} \quad \dots (\because \tan 45^\circ = 1)$$

$\therefore$ Slope of the line is 1.

2. (A) Complete the following activities and rewrite it (any two) :

[4]

(1) In the following figure, seg PS is a tangent segment, line PR is a secant. If PQ = 3.6, QR = 6.4, find PS.



Activity :

$$PS^2 = \boxed{} \times PR \quad (\text{Tangent secant segment theorem})$$

$$= PQ \times [PQ + QR]$$

$$= \boxed{} \times [3.6 + 6.4]$$

$$= 3.6 \times \boxed{}$$

$$\therefore PS^2 = 36$$

$$PS = \boxed{}$$

Ans:

$$PS^2 = \boxed{\text{PQ}} \times PR \quad (\text{Tangent secant segment theorem})$$

$$= PQ \times [PQ + QR]$$

$$= \boxed{3.6} \times [3.6 + 6.4]$$

$$= 3.6 \times \boxed{10}$$

$$\therefore PS^2 = 36$$

$$PS = \boxed{6}$$

(2) If $\sec \theta = \frac{25}{7}$, find the value of $\tan \theta$.

Activity :

$$1 + \tan^2 \theta = \boxed{}$$

$$\therefore 1 + \tan^2 \theta = \boxed{}^2$$

$$\therefore \tan^2 \theta = \frac{625}{49} - \boxed{}$$

$$= \frac{625 - 49}{49}$$

$$\therefore \tan^2 \theta = \frac{576}{49}$$

$$\therefore \tan \theta = \boxed{}$$

Ans:

$$1 + \tan^2 \theta = \boxed{\sec^2 \theta}$$

$$\therefore 1 + \tan^2 \theta = \boxed{\left(\frac{25}{7}\right)^2}$$

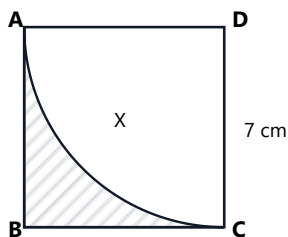
$$\therefore \tan^2 \theta = \frac{625}{49} - \boxed{1}$$

$$= \frac{625 - 49}{49}$$

$$\therefore \tan^2 \theta = \frac{576}{49}$$

$$\therefore \tan \theta = \boxed{\frac{24}{7}}$$

(3) In the following figure, side of square ABCD is 7 cm. With centre D and radius DA, sector D-AXC is drawn. Fill in the following boxes properly and find out the area of shaded region :



Activity :

$$\text{Area of square} = (\text{side})^2 = (7)^2$$

$$\therefore \text{Area of square} = \boxed{} \text{ cm}^2$$

If area of sector (D-AXC) = 38.5 cm^2 , then

$$\text{A (shaded region)} = \text{A } \boxed{} - \text{A } \boxed{}$$

$$= 49 \text{ cm}^2 - 38.5 \text{ cm}^2$$

$$= \boxed{} \text{ cm}^2$$

Ans:

$$\text{Area of square} = (\text{side})^2 = (7)^2$$

$$\therefore \text{Area of square} = \boxed{49} \text{ cm}^2$$

If area of sector (D-AXC) = 38.5 cm^2 , then

$$\text{A (shaded region)} = \text{A } \boxed{\text{(square ABCD)}} - \text{A } \boxed{\text{(sector D - AXC)}}$$

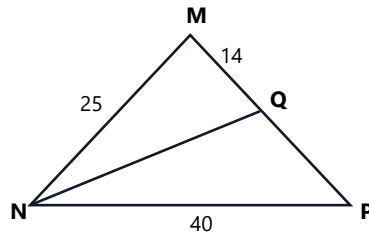
$$= 49 \text{ cm}^2 - 38.5 \text{ cm}^2$$

$$= \boxed{10.5} \text{ cm}^2$$

(B) Solve the following sub-questions (any four) :

[8]

(1) Find QP using given information in the figure :



Ans:

In $\triangle MNP$, ray NQ bisects $\angle MNP$ (Given)

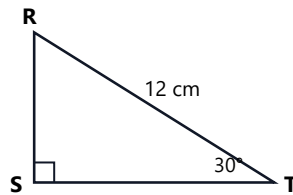
$$\frac{MQ}{QP} = \frac{NM}{NP} \quad \text{..... (By the property of angle bisector of a triangle)}$$

$$\therefore \frac{14}{QP} = \frac{25}{40}$$

$$\therefore QP = \frac{14 \times 40}{25} = \frac{560}{25}$$

$$\therefore \mathbf{QP = 22.4 \text{ units}}$$

(2) In $\triangle RST$, $\angle S = 90^\circ$, $\angle T = 30^\circ$, $RT = 12 \text{ cm}$, then find RS and ST.



Ans:

In $\triangle RST$, $\angle S = 90^\circ$, $\angle T = 30^\circ$

$\therefore \angle R = 60^\circ$ (Sum of the angles of a triangle is 180°)

By 30° - 60° - 90° theorem,

$$RS = \frac{1}{2} \times RT \quad \text{..... (Side opposite to } 30^\circ)$$

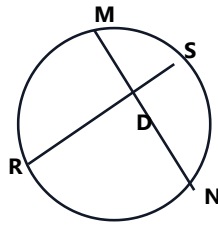
$$\therefore RS = \frac{1}{2} \times 12 = 6 \text{ cm}$$

$$ST = \frac{\sqrt{3}}{2} \times RT \quad \text{..... (Side opposite to } 60^\circ)$$

$$\therefore ST = \frac{\sqrt{3}}{2} \times 12 = 6\sqrt{3} \text{ cm}$$

$$\therefore \mathbf{RS = 6 \text{ cm} \text{ and } ST = 6\sqrt{3} \text{ cm}}$$

(3) In the following figure, chord MN and chord RS intersect at point D. If RD = 15, DS = 4, MD = 8, find DN.



Ans:

Chord MN and chord RS intersect each other at point D inside the circle.

$MD \times DN = RD \times DS$ (Theorem of internal division of chords)

$$\therefore 8 \times DN = 15 \times 4$$

$$\therefore DN = \frac{60}{8}$$

$$\therefore \mathbf{DN = 7.5 \text{ units}}$$

(4) Find the co-ordinates of the midpoint of the line segment joining points P(0, 6) and Q(12, 20).

Ans:

Let $P(0, 6) \equiv (x_1, y_1)$ and $Q(12, 20) \equiv (x_2, y_2)$

Let $M(x, y)$ be the midpoint of seg PQ.

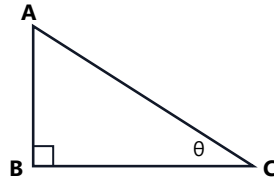
By midpoint formula,

$$x = \frac{x_1 + x_2}{2} = \frac{0 + 12}{2} = \frac{12}{2} = 6$$

$$y = \frac{y_1 + y_2}{2} = \frac{6 + 20}{2} = \frac{26}{2} = 13$$

$\therefore$ Co-ordinates of the midpoint of seg PQ are **(6, 13)**.

(5) Prove that : $\sin^2\theta + \cos^2\theta = 1$.



Ans:

In $\triangle ABC$, $\angle ABC = 90^\circ$, $\angle C = \theta^\circ$.

$$AB^2 + BC^2 = AC^2 \quad \text{..... (Pythagoras theorem)}$$

Divide both sides by AC^2

$$\frac{AB^2}{AC^2} + \frac{BC^2}{AC^2} = \frac{AC^2}{AC^2}$$

$$\therefore \left(\frac{AB}{AC}\right)^2 + \left(\frac{BC}{AC}\right)^2 = 1$$

$$\text{But } \frac{AB}{AC} = \sin \theta \text{ and } \frac{BC}{AC} = \cos \theta$$

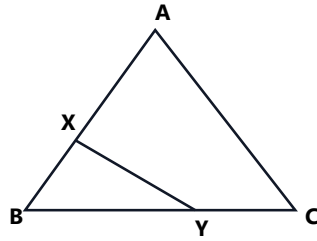
$$\therefore \sin^2\theta + \cos^2\theta = 1$$

Hence proved.

3. (A) Complete the following activities and rewrite it (any one) :

[3]

(1) In the following figure, seg XY $\parallel$ side AC. If $2AX = 3BX$ and $XY = 9$, complete the activity to find the value of AC.



Activity :

$$2AX = 3BX$$

$$\therefore \frac{AX}{BX} = \frac{3}{2}$$

$$\therefore \frac{AX + BX}{BX} = \frac{\boxed{} + \boxed{}}{2} \dots\dots\dots (\text{By componendo})$$

$$\therefore \frac{AB}{BX} = \frac{\boxed{}}{2}$$

$\triangle BCA \sim \triangle BYX$ ($\boxed{}$ Test of similarity)

$$\therefore \frac{BA}{BX} = \frac{AC}{XY} \dots\dots\dots (\text{Corresponding sides of similar triangles})$$

$$\therefore \frac{5}{\boxed{}} = \frac{AC}{9}$$

$$\therefore AC = \boxed{}$$

Ans:

$$2AX = 3BX$$

$$\therefore \frac{AX}{BX} = \frac{3}{2}$$

$$\therefore \frac{AX + BX}{BX} = \frac{\boxed{3} + \boxed{2}}{2} \dots\dots\dots (\text{By componendo})$$

$$\therefore \frac{AB}{BX} = \frac{\boxed{5}}{2}$$

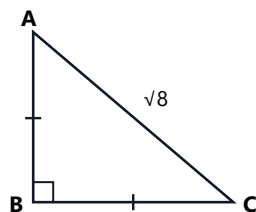
$\triangle BCA \sim \triangle BYX$ ($\boxed{AA}$ Test of similarity)

$$\therefore \frac{BA}{BX} = \frac{AC}{XY} \dots\dots\dots (\text{Corresponding sides of similar triangles})$$

$$\therefore \frac{5}{\boxed{2}} = \frac{AC}{9}$$

$$\therefore AC = \boxed{22.5}$$

(2) For finding AB and BC with the help of information given in the following figure, complete the following activity :



Activity :

$$\begin{aligned}
 AB &= BC \quad \dots\dots\dots \boxed{} \\
 \therefore \angle BAC &= \boxed{} \\
 \therefore AB &= BC = \boxed{} \times AC \\
 &= \frac{1}{\sqrt{2}} \times \boxed{} \\
 &= \frac{1}{\sqrt{2}} \times 2 \times \boxed{} \\
 AB &= BC = \boxed{}
 \end{aligned}$$

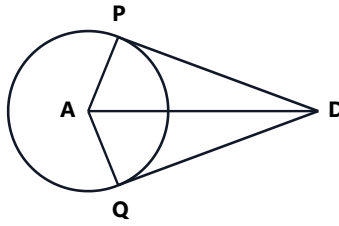
Ans:

$$\begin{aligned}
 AB &= BC \quad \dots\dots\dots \boxed{\text{(Given)}} \\
 \therefore \angle BAC &= \boxed{45^\circ} \\
 \therefore AB &= BC = \boxed{\frac{1}{\sqrt{2}}} \times AC \\
 &= \frac{1}{\sqrt{2}} \times \boxed{\sqrt{8}} \\
 &= \frac{1}{\sqrt{2}} \times 2 \times \boxed{\sqrt{2}} \\
 AB &= BC = \boxed{2}
 \end{aligned}$$

(B) Solve the following sub-questions (any two) :

[6]

(1) Prove that : The tangent segments drawn from an external point to the circle are congruent.



Ans:

Theorem : Tangent segments drawn from an external point to the circle are congruent.

To prove : $\text{seg DP} \cong \text{seg DQ}$

Construction : Draw radius AP and radius AQ.

Proof : In $\triangle PAD$ and $\triangle QAD$,

$\text{seg PA} \cong \text{seg QA}$ (radii of the same circle)

$\text{seg AD} \cong \text{seg AD}$ (Common side)

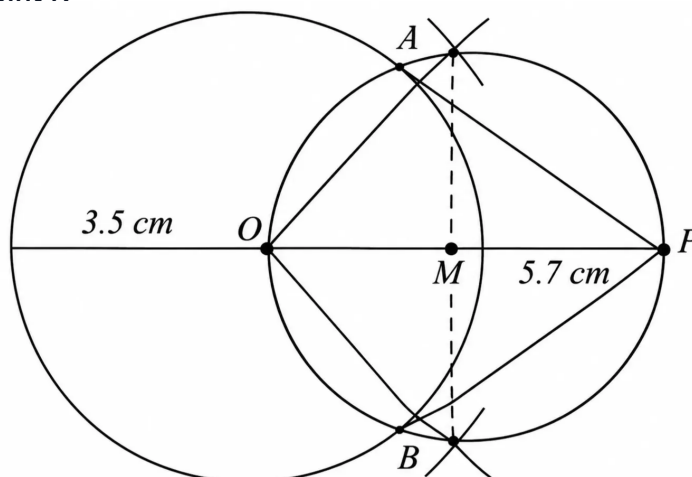
$\angle APD = \angle AQD = 90^\circ$ (Tangent theorem)

$\therefore \triangle PAD \cong \triangle QAD$ (By SAS test / Hypotenuse-side test)

$\therefore \text{seg DP} \cong \text{seg DQ}$ (corresponding sides of congruent triangles)

Hence proved.

(2) Draw a circle with centre O and radius 3.5 cm, take a point P at a distance 5.7 cm from the centre. Draw tangents to the circle from point P.



Ans:

Steps of Construction :

1. Draw a circle with centre O and radius 3.5 cm.
2. Take a point P such that $OP = 5.7$ cm.
3. Draw the perpendicular bisector of seg OP to obtain midpoint M.
4. With centre M and radius MO, draw an arc intersecting the circle at two points.
5. Draw lines passing through point P and these intersection points to obtain the required tangents.

(3) Find the co-ordinates of point P, if P divides the line segment joining the points A(-1, 7) and B(4, -3) in the ratio 2 : 3.

Ans:

Let A(-1, 7) $\equiv$ (x_1 , y_1) and B(4, -3) $\equiv$ (x_2 , y_2) and $m : n = 2 : 3$

By section formula,

$$x = \frac{mx_2 + nx_1}{m + n} = \frac{2 \times 4 + 3 \times (-1)}{2 + 3} = \frac{8 - 3}{5} = \frac{5}{5} = 1$$

$$y = \frac{my_2 + ny_1}{m + n} = \frac{2 \times (-3) + 3 \times 7}{2 + 3} = \frac{-6 + 21}{5} = \frac{15}{5} = 3$$

$\therefore$ Co-ordinates of point P are **(1, 3)**.

(4) A bucket is frustum shaped. Its height is 28 cm. Radii of circular faces are 12 cm and 15 cm. Find the capacity of the bucket in litres. ($\pi = 22/7$)

Ans:

For the frustum shaped bucket :

height $h = 28$ cm, $r_1 = 15$ cm, $r_2 = 12$ cm

$$\text{Volume of frustum} = \frac{1}{3} \pi h (r_1^2 + r_2^2 + r_1 \times r_2) \dots\dots (\text{Formula})$$

$$= \frac{1}{3} \times \frac{22}{7} \times 28 \times (15^2 + 12^2 + 15 \times 12)$$

$$= \frac{22}{3} \times 4 \times (225 + 144 + 180)$$

$$= \frac{88}{3} \times 549 = 88 \times 183$$

$$\therefore \text{Volume of frustum} = 16104 \text{ cm}^3$$

Now, 1 litre = 1000 cm^3

$$\therefore \text{Capacity of the bucket} = \frac{16104}{1000} = \mathbf{16.104 \text{ litres}}$$

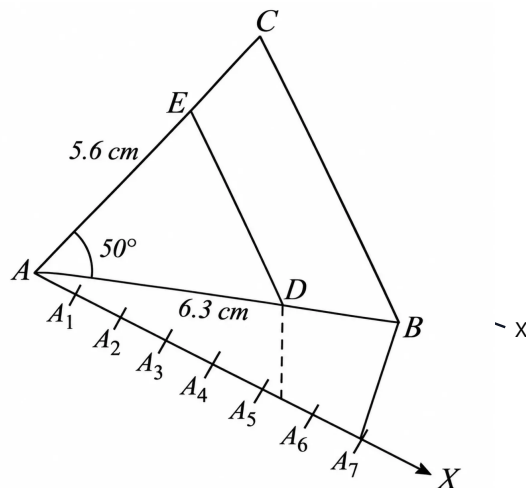
$\therefore$ The capacity of the bucket is 16.104 litres.

4. Solve the following sub-questions (any two) :

[8]

(1) $\triangle ABC \sim \triangle ADE$,

In $\triangle ABC$, $AB = 6.3$ cm, $\angle CAB = 50^\circ$, $AC = 5.6$ cm and $\frac{AB}{AD} = \frac{7}{5}$. Construct $\triangle ADE$.



Ans:

Steps of Construction :

1. Construct $\triangle ABC$ with $AB = 6.3$ cm, $\angle CAB = 50^\circ$, $AC = 5.6$ cm.
2. Draw a ray AX making an acute angle with AB on the opposite side of vertex C .
3. Mark 7 equal segments on ray AX as $A_1, A_2, \dots, A_7$.
4. Join A_7 to B . Draw a line through A_5 parallel to A_7B intersecting seg AB at point D .
5. Draw a line through D parallel to side BC intersecting seg AC at point E .

$\therefore \triangle ADE$ is the required triangle.

(2) Surface area of sphere and total surface area of cube are same. Show that the ratio of the volume of the sphere to that of the cube is $\sqrt{6} : \sqrt{\pi}$.

Ans:

Let the radius of the sphere be r and the side of the cube be a .

Surface area of sphere = Total surface area of cube (Given)

$$\therefore 4\pi r^2 = 6a^2 \text{ (I)}$$

$$\therefore \frac{r^2}{a^2} = \frac{6}{4\pi} = \frac{3}{2\pi}$$

$$\therefore \frac{r}{a} = \frac{\sqrt{3}}{\sqrt{2\pi}} \text{ (Taking square roots of both sides) ... (II)}$$

$$\text{Now, } \frac{\text{Volume of sphere}}{\text{Volume of cube}} = \frac{\frac{4}{3}\pi r^3}{a^3} = \frac{4\pi r^2 \times r}{3a^2 \times a}$$

$$= \frac{6a^2 \times r}{3a^2 \times a} \text{ [From (I), } 4\pi r^2 = 6a^2 \text{]}$$

$$= \frac{2r}{a}$$

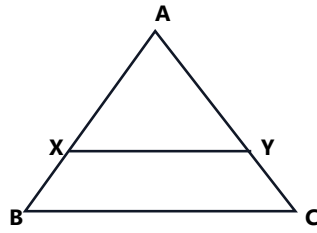
$$= 2 \times \frac{\sqrt{3}}{\sqrt{2\pi}} \text{ [From (II)]}$$

$$= \frac{\sqrt{4 \times 3}}{\sqrt{2\pi}} = \frac{\sqrt{6}}{\sqrt{\pi}}$$

$$\therefore \text{Volume of sphere : Volume of cube} = \sqrt{6} : \sqrt{\pi}$$

Hence, proved.

(3) In $\triangle ABC$, A-X-B and A-Y-C is such that segment $XY \parallel$ side BC. Segment XY divides $\triangle ABC$ in two equal areas, then find $\frac{BX}{AB}$.



Ans:

seg $XY \parallel$ side BC and XY divides $\triangle ABC$ into two parts of equal areas. (Given)

$$\therefore A(\triangle ABC) = 2 \times A(\triangle AXY) \text{ (I)}$$

In $\triangle AXY$ and $\triangle ABC$,

$$\angle AXY \cong \angle ABC \text{ (Corresponding angles, as } XY \parallel BC)$$

$$\angle XAY \cong \angle BAC \text{ (Common angle)}$$

$$\therefore \triangle AXY \sim \triangle ABC \text{ (AA test of similarity)}$$

$$\therefore \frac{A(\triangle AXY)}{A(\triangle ABC)} = \frac{AX^2}{AB^2} \text{ (Theorem of areas of similar triangles)}$$

$$\therefore \frac{A(\triangle AXY)}{2 \times A(\triangle AXY)} = \frac{AX^2}{AB^2} \text{ [From (I)]}$$

$$\therefore \frac{1}{2} = \frac{AX^2}{AB^2}$$

$$\therefore \frac{AX}{AB} = \frac{1}{\sqrt{2}} \text{ (Taking square roots of both sides) ... (II)}$$

$$\text{Now, } \frac{BX}{AB} = \frac{AB - AX}{AB} \text{ (A-X-B)}$$

$$\therefore \frac{BX}{AB} = 1 - \frac{AX}{AB}$$

$$\therefore \frac{BX}{AB} = 1 - \frac{1}{\sqrt{2}} \text{ [From (II)]}$$

$$\therefore \frac{BX}{AB} = \frac{\sqrt{2} - 1}{\sqrt{2}}$$

5. Solve the following sub-questions (any one) :

[3]

(1) Seg PQ is a chord of a circle. Arc PXQ is a major arc and Arc PYQ is a minor arc. $\angle POQ$ is central angle of the circle :

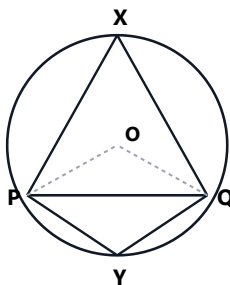
(a) Draw the figure from given information.

(b) Find the measure of $\angle PXQ$, $\angle PYQ$ and add $m\angle PXQ$, $m\angle PYQ$.

(c) Find the measure of $\angle POQ$. Write the relation between $\angle PXQ$ and $\angle POQ$.

Ans:

(a) Figure :



(b) $\angle PXQ$ is inscribed in the major arc PXQ and it intercepts arc PYQ.

$$\therefore m\angle PXQ = \frac{1}{2} m(\text{arc PYQ}) \dots\dots (\text{Inscribed angle theorem}) \dots (I)$$

Similarly, $\angle PYQ$ is inscribed in the minor arc PYQ and it intercepts arc PXQ.

$$\therefore m\angle PYQ = \frac{1}{2} m(\text{arc PXQ}) \dots\dots (\text{Inscribed angle theorem}) \dots (II)$$

Adding (I) and (II),

$$m\angle PXQ + m\angle PYQ = \frac{1}{2} [m(\text{arc PYQ}) + m(\text{arc PXQ})] = \frac{1}{2} \times 360^\circ$$

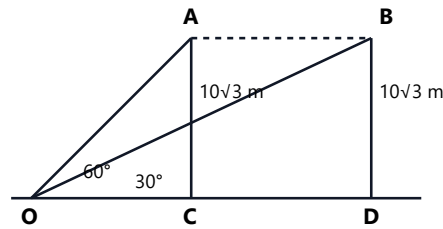
$$\therefore m\angle PXQ + m\angle PYQ = 180^\circ$$

(c) $m\angle POQ = m(\text{arc PYQ}) \dots\dots (\text{Definition of measure of minor arc})$

Relation : The measure of a central angle is twice the measure of the inscribed angle intercepting the same arc.

$$\therefore m\angle POQ = 2 m\angle PXQ$$

(2) Two eagles are flying in the sky at height of $10\sqrt{3}$ m above the ground. A boy standing at a point on the ground looks towards the eagles making angles of elevation of 60° and 30° respectively. Find the distance between these two eagles.



Ans:

Let the boy be standing at point O. Let A and B be the positions of the two eagles.

Let A and B be at heights AC and BD respectively, with C and D on the ground.

$$AC = BD = 10\sqrt{3} \text{ m}, \angle AOC = 60^\circ, \angle BOD = 30^\circ$$

$$\text{In right-angled } \triangle OCA, \quad \tan 60^\circ = \frac{AC}{OC}$$

$$\therefore \sqrt{3} = \frac{10\sqrt{3}}{OC} \quad \dots\dots (\because \tan 60^\circ = \sqrt{3})$$

$$\therefore OC = 10 \text{ m}$$

$$\text{In right-angled } \triangle ODB, \quad \tan 30^\circ = \frac{BD}{OD}$$

$$\therefore \frac{1}{\sqrt{3}} = \frac{10\sqrt{3}}{OD} \quad \dots\dots (\because \tan 30^\circ = \frac{1}{\sqrt{3}})$$

$$\therefore OD = 10\sqrt{3} \times \sqrt{3} = 30 \text{ m}$$

Both the eagles are flying at the same height.

$$\therefore AB = CD = OD - OC = 30 - 10 = 20$$

$\therefore$ The distance between the two eagles is **20 m**.